3.34 Affine Algebras

Definition 3.34.1 (Affine Domain)
*We call a finitely-generated k-algebra an **affine algebra**. If in addition it's integral we call it an **affine domain**.*

3.34.1 Reduction

Recall that $A_{\text{red}} \coloneqq A/N(A)$.

Lemma 3.34.2
Let A be a k-algebra and $\mathfrak{a}$ an ideal. Then $N(A/\mathfrak{a}) = \sqrt{\mathfrak{a}}/\mathfrak{a}$ and there is a canonical isomorphism

$$\begin{array}{ccc} A/\mathfrak{a} & \twoheadleftarrow & A \\ \downarrow & & \downarrow \\ (A/\mathfrak{a})_{\text{red}} & \xrightarrow{\sim} & A/\sqrt{\mathfrak{a}} \end{array}$$

$$\overline{a+\mathfrak{a}} \longrightarrow (a+\sqrt{\mathfrak{a}})$$

which is the unique morphism making the diagram commute.

Proof. For the first part $(a+\mathfrak{a})^n = a^n + \mathfrak{a}$, so $(a+\mathfrak{a}) \in N(A/\mathfrak{a}) \iff (a+\mathfrak{a})^n = 0 \iff a^n \in \mathfrak{a}$ some $n \iff a \in \sqrt{\mathfrak{a}}$.

The second part then follows from (3.3.58). ∎

Proposition 3.34.3
Let A be a k-algebra then for every reduced k-algebra B there is a canonical bijection

$$\begin{aligned} \text{Mor}(A_{\text{red}}, B) &\xrightarrow{\sim} \text{Mor}(A, B) \\ \phi &\to \phi \circ \pi \end{aligned}$$

where $\pi : A \to A_{\text{red}}$ is the quotient map, which is natural in B. When A is reduced then π is an isomorphism.

Proof. Suppose $\tilde{\phi} : A \to B$ is a homomorphism and $a \in N(A)$. Then $0 = \tilde{\phi}(a^n) = \tilde{\phi}(a)^n$ whence $\tilde{\phi}(N(A)) \subset N(B) = (0)$, that is $N(A) \subset \ker(\tilde{\phi})$. By (3.3.56) there exists ϕ such that $\phi \circ \pi = \tilde{\phi}$. This shows the given map is surjective. As π is surjective we may also deduce that the map is injective. ∎

Proposition 3.34.4
Let A, B be k-algebras and $\phi : A \to B$ be a k-algebra homomorphism, then there is a unique morphism $\phi_{\text{red}} : A_{\text{red}} \to B_{\text{red}}$ making the following diagram commute

$$\begin{array}{ccc} A & \xrightarrow{\phi} & B \\ \downarrow & & \downarrow \\ A_{\text{red}} & \dashrightarrow & B_{\text{red}} \end{array}$$

given by

$$\phi_{\text{red}}(a + N(A)) = \phi(a) + N(B)$$

Furthermore $(\psi \circ \phi)_{\text{red}} = \psi_{\text{red}} \circ \phi_{\text{red}}$. If ϕ is an isomorphism then so is ϕ_{red}.

Proof. In the notation of (3.34.3) ϕ_{red} is simply the morphism corresponding to $\pi_B \circ \phi$, from which existence and uniqueness follows.

Uniqueness then also shows that this is functorial. ∎

3.34.2 Normalisation

The following normalisation results can be seen as a refinement of results on transcendence bases (see the Transcendental Field Extensions subsection).

Definition 3.34.5 (Algebraically Independent)
*Let A be a k-algebra and $x_1, ..., x_n$ elements of A. Then we say they are **algebraically independent** if one of the following equivalent conditions holds*

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- *The unique k-algebra homomorphism $\phi : k[X_1, ..., X_n] \to A$ such that $\phi(X_i) = x_i$ (evaluation homomorphism) is injective*
- *There are no non-zero polynomials $f(X_1, ..., X_n)$ such that $f(x_1, ..., x_n) = 0$.*

Note in particular it induces an isomorphism $k[X_1, ..., X_n] \xrightarrow{\sim} k[x_1, ..., x_n] \subset A$.

Definition 3.34.6 (Normalising Family)
*Let A be a finitely-generated k-algebra. A **normalising family** is a set $\{x_1, ..., x_n\}$ of elements of A such that*

- *$x_1, ..., x_n$ are **algebraically independent** over k*
- *A is a finite $k[x_1, ..., x_n]$-module (equivalently integral over $k[x_1, ..., x_n]$).*

NB this is completely equivalent to specifying an integral, injective map

$$k[X_1, ..., X_n] \hookrightarrow A$$

This may be seen as a refined transcendence base. More precisely we have the following

Proposition 3.34.7 (Relationship to Transcendence Base)
Let A be an integral finitely-generated k-algebra with $K := \mathrm{Frac}(A)$. Let $S \subset A$ be a subset. Then

- *If A is integral over $k[S]$ then $K/k(S)$ is algebraic*
- *If S is a normalising family for A then S is a transcendence basis for K/k*

In particular normalising families have order $\mathrm{trdeg}(K/k)$.

Proof. By (3.18.45) the set $\{x \in K \mid x \text{ algebraic over } k(S)\}$ forms a subfield containing A, and therefore equals K.

The final statement follows from (3.18.141). ∎

The following is useful as it removes the necessity of showing algebraic independence in certain cases.

Corollary 3.34.8
Let A be an integral finitely-generated k-algebra with $K := \mathrm{Frac}(A)$. Let $S \subset A$ be a subset such that

- *A is integral over $k[S]$*
- *$\#S \leq \mathrm{trdeg}_k(K)$*

*then S is a **normalising family**.*

Proof. This follows from the previous result and (3.18.141). ∎

There are a few forms of the Normalisation Lemma which we prove, of progressively stronger form.

Lemma 3.34.9 (Hypersurface Normalisation Lemma)
Let $A = k[X_1, ..., X_n]$ be a polynomial ring over an infinite field k and $0 \neq F \in A$. Then there exists $\lambda_1, ..., \lambda_{n-1} \in k$ such that

- *$x_i := X_i - \lambda_i F \quad 1 \leq i \leq n-1$*
- *$x_1, ..., x_{n-1}, F$ is a normalising family for A*
- *$FA \cap k[x_1, ..., x_{n-1}, F] = Fk[x_1, ..., x_{n-1}, F]$*

This may be viewed as a commutative diagram

$$\begin{array}{ccc} \phi : k[Y_1, ..., Y_n] & \xrightarrow{\sim} & k[X_1, ..., X_n] \\ \uparrow & & \downarrow \\ k[Y_1, ..., Y_{n-1}] & \hookrightarrow & k[X_1, ..., X_n]/(F) \end{array}$$

where the horizontal arrows are injective, integral (and finite) and $\phi^{-1}((F)) = (Y_n)$.

Remark 3.34.10
Reversing the arrows we get the geometric picture, where horizontal arrows are finite and surjective

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$$\begin{array}{ccc} \mathbb{A}^n & \xrightarrow{\sim} & \mathbb{A}^n \\ \uparrow & & \downarrow \\ V(F) & \twoheadrightarrow & \mathbb{A}^{n-1} \end{array}$$

in other words after a linear change of variables we may express $V(F)$ as a finite covering of a standard hyperplane.

Proposition 3.34.11 (Nagata Normalisation Lemma)
*Let $A = k[x_1, ..., x_n]$ be a finitely-generated k-algebra such that k is infinite. Then there exists a **normalising family** $y_1, ..., y_d \in A$ such that each y_i is a k-linear combination of $x_1, ..., x_n$. Further if $\mathfrak{a}_1 \lhd A$ is a proper ideal, then these may be chosen such that*

$$\mathfrak{a}_1 \cap k[y_1, ..., y_d] = (y_1, ..., y_h)k[y_1, ..., y_d]$$

for some $0 \leq h \leq d$, where $h = 0$ denotes the zero ideal.

Proposition 3.34.12 (Bourbaki Normalisation Lemma)
*Let $A = k[x_1, ..., x_n]$ be a finitely-generated k-algebra such that k is infinite. Then there exists a **normalising family** $y_1, ..., y_d \in A$ such that each y_i is a k-linear combination of $x_1, ..., x_n$.*

Furthermore for any finite chain of proper ideals in A

$$\mathfrak{a}_1 \subseteq \mathfrak{a}_2 \subseteq ... \subseteq \mathfrak{a}_p \subsetneq A$$

the family may be chosen such that

$$\mathfrak{a}_j \cap k[y_1, ..., y_d] = \left(y_1, ..., y_{h(j)}\right)k[y_1, ..., y_d] \quad 1 \leq j \leq p \ .$$

for some non-decreasing sequence of integers $h(j)$, where $h(j) = 0$ denotes the zero ideal.

Remark 3.34.13 (Geometric Interpretation)
Note the normalisation here is equivalent to a commutative diagram

$$\begin{array}{ccc} k[Y_1, ..., Y_d] & \hookrightarrow & A \\ \uparrow & & \downarrow \\ k\left[Y_{h(1)+1}, ..., Y_d\right] & \hookrightarrow & A/\mathfrak{a}_1 \\ \uparrow & & \downarrow \\ \vdots & & \vdots \\ \uparrow & & \downarrow \\ k\left[Y_{h(p)+1}, ..., Y_d\right] & \hookrightarrow & A/\mathfrak{a}_p \end{array}$$

where the horizontal arrows are integral and injective, and the top arrow is given by

$$\phi : Y_i \to \sum_j \lambda_{ij} x_j$$

such that $\phi^{-1}(\mathfrak{a}_i) = \left(Y_1, ..., Y_{h(i)}\right)$

As before this expresses A as a finite covering of $\mathbb{A}^d$ under which each subvariety is also a finite covering of a standard linear subspace.

We first prove the weaker form of the Normalisation Lemma

Proof of (3.34.9). First decompose F into homogenous polynomials (monomials of the same degree)

$$F = F_0 + F_1 + ... + F_m$$

and observe that the monomial X_n^m appears only in F_m. Define

$$F' \coloneqq F(X_1 + \lambda_1 X_n, ..., X_{n-1} + \lambda_{n-1} X_n, X_n)$$

Furthermore in terms of homogenous polynomials

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$$F'_m = F_m(X_1 + \lambda_1 T, ..., X_{n-1} + \lambda_{n-1} X_n, X_n)$$

and the coefficient of X_n^m in F' is simply $F'_m(0, ..., 0, 1) = F_m(\lambda_1, ..., \lambda_{n-1}, 1) \in k$. There are only finitely many values of λ such that F_m is zero, whence there exists a λ such that X_n^m has non-zero coefficient in F', whence F' is monic. By the previous Lemma we have $F'(x_1, ..., x_{n-1}, X_n) = 0$, with the leading coefficient in X_n constant.

Let $B \coloneqq k[x_1, ...x_{n-1}, F]$. Then clearly $B[X_n] = A$ and X_n is integral over B. Therefore by (3.23.4) A is a finite B-module. As $\operatorname{trdeg}(A/k) = n$ by (3.34.8) $\{x_1, ..., x_{n-1}, F\}$ is a normalising family, and B is isomorphic to a polynomial ring.

As B is a polynomial ring it is integrally closed (3.23.12). Then the final statement is a consequence of the following lemma (3.34.14) (essentially to prove that $V(F) \to \mathbb{A}^{n-1}$ is surjective). ∎

Lemma 3.34.14
Let A be integrally closed, $\phi : A \hookrightarrow B$ injective and integral and $a \in A$. Then

$$(a)^{\mathrm{ec}} = \phi^{-1}(\phi(a)B) = (a)$$

Proof. Let $K = \operatorname{Frac}(A)$ and $L = \operatorname{Frac}(B)$, then ϕ extends to an injection $\phi : K \hookrightarrow L$.

Generically we have $(a) \subseteq (a)^{\mathrm{ec}}$. Suppose $a' \in (a)^{\mathrm{ec}}$, then $\phi(a') = \phi(a)b$. Therefore $\phi(a'a^{-1}) = b$ is integral over A, whence so is $a'a^{-1}$. As A is integrally closed we have $a' \in (a)$ as required. ∎

Before proving the stronger version of Normalisation Lemma, we need some preliminary technical results

Lemma 3.34.15
Let A be a k-algebra and $\mathfrak{a} \lhd A$ an ideal. Then $\mathfrak{a}$ is proper $\iff \mathfrak{a} \cap k = \{0\}$.

Lemma 3.34.16
Let $A = k[X_1, ..., X_n]$ be a polynomial ring and $\mathfrak{a} \lhd A$ a proper ideal. Then TFAE

a) $\mathfrak{a} = (X_1, ..., X_h)$
b)
  i) $X_1, ..., X_h \in \mathfrak{a}$
  ii) $\mathfrak{a} \cap k[X_{h+1}, ..., X_n] = \{0\}$

In this case $\mathfrak{a} \cap k[X_1, ..., X_h] = (X_1, ..., X_h)k[X_1, ..., X_h]$

Proof. We claim that there is a direct sum of k-vector spaces

$$k[X_1, ..., X_n] = (X_1, ..., X_h) \oplus k[X_{h+1}, ..., X_n]$$

from which the result largely follows. Let S be the set of monomials in which at least one of $X_1, ..., X_h$ appears, and let T be the set for which none appears (but including 1). Then clearly $A = \langle S \rangle \oplus \langle T \rangle$ and $k[X_{h+1}, ..., X_n] = \langle T \rangle$. We argue that $(X_1, ..., X_h) = \langle S \rangle$. First S is stable under multiplication by $X_1, ..., X_n$, and so $\langle S \rangle$ is an ideal. One inclusion is obvious, furthermore it's clear that $S \subseteq (X_1, ..., X_n)$ from which the claim follows. ∎

We may now proceed to the proof of the stronger versions of the Normalisation Lemma.

Reduction to polynomial ring case for (3.34.11), (3.34.12). We show that for both forms it is possible to reduce to the case of a polynomial ring. For let A be a finitely-generated k-algebra then we may write $A \coloneqq k[X_1, ..., X_n]/\mathfrak{a}$ for some ideal $\mathfrak{a}$. Then the polynomial ring case for $p = 1$ shows the existence of an integral, injective map

$$\phi : k[Y_1, ..., Y_m] \hookrightarrow A$$

If $\mathfrak{a}_i$ is a chain of ideals in A, then $\mathfrak{a}'_i \coloneqq \phi^{-1}(\mathfrak{a}_i)$ is a chain of ideals in $k[Y_1, ..., Y_m]$. The general case for a polynomial ring shows the existence of an integral map

$$\psi : k[Z_1, ..., Z_m] \hookrightarrow k[Y_1, ..., Y_m]$$

such that

$$\psi^{-1}(\mathfrak{a}'_i) = \left(Z_1, ..., Z_{h(i)}\right)$$

The composition $\phi \circ \psi$ gives the required normalisation of A. Geometrically express A as a finite covering of affine space, by considering it as a subvariety of larger affine space. ∎

Proof of (3.34.11) in the polynomial ring case. Let $A = k[X_1, ..., X_n]$ and proceed by induction on n.

Note that as $\mathfrak{a}_1$ is proper, we must have $\mathfrak{a}_1 \cap k = \{0\}$, and we may obviously also assume that $\mathfrak{a}_1 \neq (0)$ (as otherwise we may take $h = 0$).

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Choose $0 \neq x_1 \in \mathfrak{a}_1$. Then by (3.34.9) there exists $t_2, ..., t_n \in A$ such that $x_1, t_2, ..., t_n$ is a normalising family and $(x_1) \cap B = x_1 B$ where $B := k[x_1, t_2, ..., t_n]$. In the case that $\mathfrak{a}_1$ is principal we are done, since the choice of x_1 was arbitrary, and in this case $h(1) = 1$. In particular this covers the base case $n = 1$ because A is a PID (3.15.4).

Otherwise $B' := k[t_2, ..., t_n]$ is a polynomial ring and $\mathfrak{a}_1' := \mathfrak{a}_1 \cap B'$ is proper by (3.34.15). By induction on n there is a normalising family $x_2, ..., x_n$ for B' such that $\mathfrak{a}_1' \cap C' = (x_2, ..., x_h)C'$ where $C' := k[x_2, ..., x_n]$ and B' is integral over C'.

Define $C := k[x_1, ..., x_n] = C'[x_1]$ then $x_1, t_2, ..., t_n$ are integral over C, so B is integral over C (3.23.4), and A is integral over C (3.23.5), so by (3.34.8) $x_1, ..., x_n$ is a normalising family for A.

We claim that $\mathfrak{a}_1'' := \mathfrak{a}_1 \cap C = (x_1, ..., x_h)C$. Clearly $x_1, ..., x_h \in \mathfrak{a}_1''$, and $\mathfrak{a}_1'' \cap k[x_{h+1}, ..., x_n] = \mathfrak{a}_1' \cap k[x_{h+1}, ..., x_n] = \{0\}$ by (3.34.16) applied to the ring C'. Then (3.34.16) applied to the ring C demonstrates the claim. ∎

Proof of (3.34.12) in the polynomial ring case. We can then show the case $p > 1$ by induction, for by the induction hypothesis there exists a normalising family $t_1, ..., t_n$ for A such that

$$\mathfrak{a}_j \cap B = \left(t_1, ..., t_{h(j)}\right)B \quad 1 \leq j \leq p-1$$
$$B := k[t_1, ..., t_n]$$

Let $r = h(p-1)$, then by the case $p = 1$ applied to the ring $B' := k[t_{r+1}, ..., t_n]$ and ideal $\mathfrak{a}_p \cap B'$ there exists a normalising family $x_{r+1}, ..., x_n$ for B' such that for some $s \leq n$ (possibly equal to r to denote the zero ideal),

$$\mathfrak{a}_p \cap C' = (x_{r+1}, ..., x_s)C'$$
$$C' := k[x_{r+1}, ..., x_n]$$

We claim that $t_1, ..., t_r, x_{r+1}, ..., x_n$ is a suitable normalising family for A, with $h(p) = s$.

For define $C := k[t_1, ..., t_r, x_{r+1}, ..., x_n] = C'[t_1, ..., t_r]$. Recall B' is integral over C', and $t_1, ..., t_r$ are obviously integral over C so $B = B'[t_1, ..., t_r]$ is integral over C by (3.23.7). Then A is integral over C by (3.23.5), and this is a normalising family by (3.34.8), and in particular algebraically independent.

For $j \leq p-1$ and $h := h(j) \leq r$, apply (3.34.16) to the ideal $\mathfrak{a}_j \cap B$ to see $\mathfrak{a}_j \cap k[t_{h+1}, ..., t_n] = \{0\}$ and therefore $\mathfrak{a}_j \cap k[t_{h+1}, ..., t_r, x_{r+1}, ..., x_n] = \{0\}$. As $t_1, ..., t_h \in \mathfrak{a}_j$ we see by (3.34.16) that $\mathfrak{a}_j \cap C = (t_1, ..., t_h)C$ as required.

Similarly by (3.34.16) $\mathfrak{a}_p \cap k[x_{s+1}, ..., x_n] = \{0\}$ and clearly $t_1, ..., t_r, x_{r+1}, ..., x_s \in \mathfrak{a}_p$. Then by (3.34.16) again $\mathfrak{a}_p \cap C = (t_1, ..., t_r, x_{r+1}, ..., x_s)C$ as required. ∎

Remark 3.34.17
In Bourbaki's proof the reduction to the polynomial ring case increases p to $p+1$, so in particular the $p = 1$ case requires the more complex reduction argument at the end of the proof. Here we use a simplified argument from [Sha00].

3.34.3 Nullstellensatz

Definition 3.34.18 (Zeros of an ideal)
*Let $\mathfrak{a} \lhd k[X_1, ..., X_n]$ be an ideal and K/k a field extension. Then a point $x \in K^n$ is a **zero** of $\mathfrak{a}$ if*

$$f \in \mathfrak{a} \implies f(x) = 0$$

*We define the **residue field** to be*

$$k(x) := k(x_1, ..., x_n)$$

and also denote

$$k[x] := k[x_1, ..., x_n]$$

*We say it is **algebraic** if $k(x)/k$ is algebraic (necessarily finite).*

The following observation is useful

Proposition 3.34.19 (Zeros are homomorphisms)
Let $\mathfrak{a} \lhd k[X_1, ..., X_n]$ be an ideal then there is a bijection

$$\mathrm{AlgHom}_k(k[X_1, ..., X_n]/\mathfrak{a}, K) \longleftrightarrow \{\text{zeros of } \mathfrak{a} \text{ in } K^n\}$$
$$\phi \longrightarrow \left(\phi\left(\overline{X}_1\right), ..., \phi\left(\overline{X}_n\right)\right)$$

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The following notion is useful in future

Definition 3.34.20 (Generic Point)
Let $\mathfrak{a} \triangleleft k[X_1, ..., X_n]$ be a **prime** ideal. then a point $\xi \in L^n$ is a **generic point** of $\mathfrak{a}$ if $\ker(\mathrm{ev}_\xi) = \mathfrak{a}$. Note one always exists, because we may take $L = \mathrm{Frac}(A)$ and $\xi_i = \overline{X}_i$.

When x is another zero of $\mathfrak{a}$ this induces a k-algebra homomorphism

$$k[\xi] \to k[x]$$

which is an isomorphism precisely when x is a generic point.

Generally we are interested in the relationship between ideals of $k[X_1, ..., X_n]$ and corresponding zeros in an extension field K/k. The following proposition is fundamental

Proposition 3.34.21 (Correspondence between ideals and zeros)
Let K/k be a field extension and $x \in K^n$. Define $\mathfrak{m}_x$ to be the kernel of the homomorphism

$$\mathrm{ev}_x : k[X_1, ..., X_n] \to K$$

Then

- *$\mathfrak{m}_x$ is a prime ideal*
- *If K/k is an algebraically closed field of transcendence degree $\geq n$ then every prime ideal is of this form.*
- *If x_i are algebraic over k (e.g. if K/k is algebraic) then $\mathfrak{m}_x$ is maximal*

In this case we have a canonical isomorphism

$$k[X_1, ..., X_n]/\mathfrak{m}_x \xrightarrow{\sim} k[x] \subset K$$

and when each x_i is algebraic then $k[x] = k(x)$ is a finite extension of k.

Proof. The canonical isomorphism follows from (3.10.3). Any subring of a field is an integral domain, which means $\mathfrak{p}_x$ is prime by (3.3.59).

By (3.18.44) x_i are algebraic over k if and only if $k(x_1, ..., x_n)/k$ is algebraic (and even finite). By the same result $k[x_1, ..., x_n] = k(x_1, ..., x_n)$ and therefore $\mathfrak{m}_x$ is maximal by (3.3.59).

Let $\mathfrak{p}$ be a prime ideal and define $k(x) \coloneqq \mathrm{Frac}(k[x])$ and $k[x] \coloneqq k[X_1, ..., X_n]/\mathfrak{p}$. If K has transcendence degree $\geq n$ then there is an embedding $k(x)/k \to K/k$ by (3.18.61). This restricts to an isomorphism $k[x] \xrightarrow{\sim} k[\overline{x}]$ for some $\overline{x}_i \in K$. It's clear that $\mathfrak{p} = \mathfrak{m}_x$. ∎

We may show that all maximal ideals correspond to zero-loci.

Lemma 3.34.22 (Zariski's Lemma)
Let k be an infinite field and A a finitely-generated k-algebra which is a field. Then A is a finite extension of k.

Proof. By Normalisation Lemma (3.34.11) there is an injective, integral map $k[X_1, ..., X_d] \hookrightarrow A$ for some $d \geq 0$. By (3.23.14) then $k[X_1, ..., X_d]$ is a field which implies $d = 0$. We conclude that A is algebraic over k, and by (3.18.44) finite over k. ∎

Proposition 3.34.23 (Weak Nullstellensatz I)
Every maximal ideal of $k[X_1, ..., X_n]/\mathfrak{a}$ is of the form $\mathfrak{m}_x/\mathfrak{a}$ for $x \in \overline{k}^n$ a zero of $\mathfrak{a}$.

In addition the conclusion of Zariski's Lemma holds for all (possibly finite) coefficient fields k.

Proof. Observe that $x \in \overline{k}^n$ is a zero of $\mathfrak{a}$ if and only if $\mathfrak{a} \subset \mathfrak{m}_x$. Therefore we may reduce to the case $\mathfrak{a} = (0)$ by results on quotient rings (3.3.56).

We've already shown that $\mathfrak{m}_x$ is maximal when $x \in \overline{k}^n$ is algebraic (3.34.21). Conversely if $\mathfrak{m}$ is a maximal ideal, then $\mathfrak{m}\overline{k}[X_1, ..., X_n]$ is a proper ideal contained in some maximal ideal $\mathfrak{m}' \triangleleft \overline{k}[X_1, ..., X_n]$. If $\mathfrak{m}'$ has an algebraic zero then so does $\mathfrak{m}$ and $\mathfrak{m} \subset \mathfrak{m}_x$ for some x, which are equal by maximality.

However $\overline{k}[X_1, ..., X_n]/\mathfrak{m}'$ is a finitely-generated $\overline{k}$-algebra which is a field and $\overline{k}$ is infinite. Therefore by (3.34.22) it is $\overline{k}$, and this yields an algebraic zero of $\mathfrak{m}'$ as required.

We may generalise Zariski's Lemma to the case k is not necessarily infinite. If A is a finitely-generated k-algebra which is a field then we have shown it is isomorphic to $k[X_1, ..., X_n]/\mathfrak{m}_x$ for some algebraic point x. By (3.34.21) is a finite extension of k, as required. ∎

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We may make the correspondence between maximal ideals and algebraic points more precise

Proposition 3.34.24 (Weak Nullstellensatz II)
There is a bijective map

$$\begin{aligned} \overline{k}^n / \operatorname{Aut}\left(\overline{k}/k\right) &\longrightarrow \{\mathfrak{m} \lhd k[X_1, ..., X_n] \text{ maximal}\} \\ x &\longrightarrow \mathfrak{m}_x \end{aligned}$$

When $x \in k^n$ then

$$\mathfrak{m}_x = (X_1 - x_1, ..., X_n - x_n)$$

Proof. The map is surjective by (3.34.23). It's well-defined because $\mathfrak{m}_{\sigma(x)} = \mathfrak{m}_x$.

By (3.34.21) we have an isomorphism $k[X_1, ..., X_n]/\mathfrak{m}_x \xrightarrow{\sim} k[x_1, ..., x_n] \subset \overline{k}$. If $\mathfrak{m}_x = \mathfrak{m}_y$ then these compose to yield an isomorphism $\sigma : k[x_1, ..., x_n] \xrightarrow{\sim} k[y_1, ..., y_n] \subset \overline{k}$ such that $\sigma(x_i) = y_i$. By (3.18.69) this extends to $\sigma \in \operatorname{Aut}\left(\overline{k}/k\right)$. Therefore the given mapping is injective. ∎

3.34.4 Morphisms of Affine Algebras

Definition 3.34.25
*Let B be a k-algebra. We say that B is **algebraic** over k if for every maximal ideal $\mathfrak{m} \lhd B$ we have $B/\mathfrak{m} \xrightarrow{\sim} k(\mathfrak{m})$ is algebraic over k.*

Proposition 3.34.26
Let B be a k-algebra, then it is algebraic in each of the following cases

a) B is a finitely generated k-algebra
b) B is a local k-algebra such that $k(\mathfrak{m}_B)/k$ is algebraic
c) B is an algebraic field extension of k
d) B is the localisation of a finitely-generated k-algebra at a maximal ideal

Proof. We prove each in turn

a) This follows from Zariski's Lemma in the general case (3.34.23)
b) The only maximal ideal is $\mathfrak{m}_B$ so this is obvious
c) This is a special case of b)
d) This follows from a) and b) because $B_{\mathfrak{m}}$ is a local ring with residue field isomorphic to $B/\mathfrak{m}$ (3.5.43).

∎

Proposition 3.34.27
Let $\phi : A \to B$ be a k-algebra homomorphism, with B algebraic. Then the inverse image of a maximal ideal is again maximal.

Proof. Let $\mathfrak{n}$ be a maximal ideal. By assumption $B/\mathfrak{n}$ is integral over k, and so a-fortiori $B/\mathfrak{n}$ is integral over A. Therefore $\phi^{-1}(\mathfrak{n})$ is maximal by (3.23.16). ∎

3.34.5 Krull Dimension of Affine Algebras

The Krull Dimension of affine domains is particularly well-behaved. Specifically they are biequidimensional and therefore satisfy a co-dimension formula (3.34.35). Further there's a geometric proof of the "Hauptidealsatz" (3.34.32). We first show that it is equal to transcendence degree in the integral case.

Proposition 3.34.28
Let A be an affine algebra with normalising family $x_1, ..., x_n$, then $\dim A = n$.

In particular every affine domain A satisfies $\dim A = \operatorname{trdeg}(\operatorname{Frac}(A)/k)$, and the polynomial ring $k[X_1, ..., X_n]$ has dimension n.

Proof. By definition A is integral over $k[x_1, ..., x_n]$, which is isomorphic to a polynomial ring so we may reduce to the case of polynomial ring by (3.23.22).

$\dim A \geq n$). This is clear by considering the chain of prime ideals

$$(X_1) \subsetneq (X_1, X_2) \subsetneq ... \subsetneq (X_1, ..., X_n)$$

$\dim A \leq n$). We may argue by the Strong Normalisation Lemma (3.34.12) and the subsequent remark that any chain of prime ideals must have length at most n, as any normalising family has order n.

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Alternatively we may proceed by induction on n to show $\dim k[X_1, ..., X_n] = n$. Consider a maximal chain

$$\mathfrak{p}_0 \subsetneq \mathfrak{p}_1 \subsetneq ... \subsetneq \mathfrak{p}_m$$

Clearly $\mathfrak{p}_0 = 0$, and $\mathfrak{p}_1 = (f)$ principal by (3.21.5). By (3.34.9) there is an integral, injective map

$$k[Y_1, ..., Y_{n-1}] \hookrightarrow k[X_1, ..., X_n]/(f)$$

whence $\dim(\mathfrak{p}_1) = \dim(k[X_1, ..., X_n]/(f)) = \dim(k[Y_1, ..., Y_{n-1}]) = n - 1$, and by definition $m - 1 \leq n - 1$. As the maximal chain was arbitrarily chosen, we have $\dim k[X_1, ..., X_n] \leq n$. The reverse inequality was already shown so we are done.

The final statement follows from the existence of a normalising family (3.34.11) and (3.34.7) ∎

Corollary 3.34.29
Let A be a finitely-generated k-algebra. Then $\dim A[T] = \dim A + 1$.

Proof. By (3.34.11) there is an injective integral ring homomorphism

$$\phi : k[X_1, ..., X_n] \hookrightarrow A$$

where $\dim A = n$. By (3.7.6) there is a ring homomorphism

$$k[X_1, ..., X_n][T] \hookrightarrow A[T]$$

which is injective. Further by assumption $A = k[y_1, ..., y_r]$ so $A[T] = k[y_1, ..., y_r, T]$ and by (3.23.4) the ring homomorphism is integral. This shows that $\phi(x_1), ..., \phi(x_n), T$ is a normalising family and the result follows from (3.34.28). ∎

Corollary 3.34.30
The ideal $(X_1, ..., X_r) \triangleleft k[X_1, ..., X_n]$ has dimension $n - r$.

Corollary 3.34.31
Let A be an integral finitely-generated k-algebra and $0 \neq f$ then $\dim A = \dim A_f$

The following proof is due to Tate, and presented in the Red Book [Mum99] (I.7 Theorem 2).

Proposition 3.34.32 (Hypersurface has pure codimension 1)
Let A be an affine domain of dimension n and $0 \neq f \in A$. Then

$$\dim((f)) = n - 1$$

$$\mathrm{ht}((f)) = 1$$

More precisely if $\mathfrak{p}$ is minimal over (f) then it has dimension $n - 1$ and height 1.

Proof. Consider the case $A = k[X_1, ..., X_n]$. Suppose first that f is prime, then $\mathfrak{p} = (f)$ and by (3.34.9) there is an integral injective map

$$k[Y_1, ..., Y_{n-1}] \hookrightarrow A/(f)$$

Therefore

$$\dim((f)) = \dim(A/(f)) \overset{(3.23.22)}{=} \dim(k[Y_1, ..., Y_{n-1}]) \overset{(3.34.28)}{=} n - 1$$

When f is not prime then, as $k[X_1, ..., X_n]$ is a UFD, we have a prime factorisation

$$f = \prod_{i=1}^{n} f_i^{m_i}$$

and the minimal prime decomposition is

$$\sqrt{(f)} = (f_1) \cap ... \cap (f_n)$$

In particular any prime minimal over (f) has the form $\mathfrak{p} = (f_i)$ for some i, and we may reduce to the case of f prime.

For an arbitrary k-algebra A we have a decomposition into minimal primes of (f)

$$\sqrt{(f)} = \mathfrak{p}_1 \cap ... \cap \mathfrak{p}_n$$

and without loss of generality $\mathfrak{p} = \mathfrak{p}_1$. We may localize to the case of a single prime, for choose $g \notin \mathfrak{p}$ and $g \in \mathfrak{p}_i$ for $i = 2...n$. Consider the localization $A \to A_g$, then we claim that

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$$\sqrt{(f/1)} = \mathfrak{p}A_g$$

For by (3.5.19) there is a correspondence between primes of A_g containing $(f/1)$ and primes of A containing f and not g, which are precisely the primes containing $\mathfrak{p}$ by (3.3.40) or (3.3.44). Therefore $\mathfrak{p}A_g$ is the only minimal prime of A_g containing $(f/1)$ and the claim follows from (3.3.46).

Note that $\dim A = \dim A_g$ as they have the same quotient field and therefore transcendence degree. Similarly $\dim(A/\mathfrak{p}) = \dim\big((A/\mathfrak{p})_{\bar{g}}\big) = \dim\big(A_g/\mathfrak{p}_g\big) = \dim\big(\mathfrak{p}A_g\big)$. So we may assume without loss of generality that $n = 1$ and $\mathfrak{p} = \sqrt{(f)}$.

By (3.34.11) there is an integral, injective map

$$\phi : B \hookrightarrow A$$

where $B = k[X_1, ..., X_n]$, which induces an algebraic field extension

$$K := \mathrm{Frac}(B) \hookrightarrow \mathrm{Frac}(A) =: L$$

We claim that there exists $f_0 \in B$ such that

$$\phi^{-1}\big(\sqrt{f}\big) = \sqrt{(f_0)}$$

Observe that in this case $\sqrt{(f_0)}$ is prime and therefore the unique minimal prime over (f_0). Therefore the result would follow from the first part and (3.23.23). Firstly for any $g \in A$ we have (3.23.18)

$$\mathrm{Norm}_{L/K}(g) \in B \cap \phi^{-1}((g))$$

(because B is a UFD and therefore integrally closed (3.23.12)). Define $f_0 := \mathrm{Norm}_{L/K}(f)$ then we see that $f_0 \in \phi^{-1}((f)) \implies \sqrt{(f_0)} \subseteq \phi^{-1}\big(\sqrt{(f)}\big)$. Conversely if $\phi(g)^n = hf$ then $g^{n[L:K]} = \mathrm{Norm}(\phi(g)^n) = \mathrm{Norm}(h)f_0 \in (f_0) \implies g \in \sqrt{(f_0)}$. Therefore the reverse inclusion holds.

Finally by the codimension 1 formula (3.21.8) we have $\mathrm{ht}((f)) = 1$. ∎

Remark 3.34.33
The argument is slightly less awkward in geometric language. Decompose into irreducibles

$$V(f) = Z_1 \cup ... \cup Z_n$$

choose a principal open affine $D(g)$ which meets only Z_1 then $Z_1 \cap D(g) = V(f) \cap D(g) = V(f/1)$ is an irreducible component of $D(g)$. We argue that $\dim(X) = \dim(D(g))$ and $\dim(Z_1) = \dim(D(g) \cap Z_1)$. Further construct finite coverings

$$V(f/1) \to V(f_0) \to H$$

onto a hyperplane in $\mathbb{A}^n$.

This allows is to prove a converse to (3.21.8)

Corollary 3.34.34 (Height 1 formula)
Let A be an affine domain. Then for a prime ideal $\mathfrak{p}$

$$\mathrm{ht}(\mathfrak{p}) = 1 \implies \dim(\mathfrak{p}) = \dim(A) - 1$$

More generally if A is an affine algebra and $\mathfrak{p}_1 \subsetneq \mathfrak{p}_2$ is a saturated chain of prime ideals then

$$\dim(\mathfrak{p}_2) = \dim(\mathfrak{p}_1) - 1$$

Proof. Choose $0 \neq f \in \mathfrak{p}$ then it follows from the previous result (3.34.32). Alternatively by (3.34.11) there is an integral injective map

$$\phi : k[X_1, ..., X_n] \hookrightarrow A$$

with $\mathfrak{q} := \phi^{-1}(\mathfrak{p})$ and $n = \dim A$. Then by Going Down we have $\mathrm{ht}(\mathfrak{q}) = 1$. By (3.21.5) $\mathfrak{q}$ is principal and by (3.34.9) there is an integral injective map

$$k[Y_1, ..., Y_{n-1}] \hookrightarrow k[X_1, ..., X_n]/\mathfrak{q}$$

therefore $\dim(\mathfrak{q}) = n - 1$. By (3.23.23) this equals $\dim(\mathfrak{p})$ and we are done.

For the second statement we may consider the affine domain $A/\mathfrak{p}_1$ and observe that $\mathrm{ht}(\mathfrak{p}_2/\mathfrak{p}_1) = 1$. Therefore

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$$\dim(\mathfrak{p}_2) = \dim(\mathfrak{p}_2/\mathfrak{p}_1) = \dim(A/\mathfrak{p}_1) - 1 = \dim(\mathfrak{p}_1) - 1$$

■

Corollary 3.34.35 (Biequidimensionality)
*Let A be an affine algebra. Then it is **quasi-biequidimensional** and satisfies the **codimension formulae** for $\mathfrak{p} \subset \mathfrak{a}$*

$$\dim \mathfrak{a} = \dim \mathfrak{p} + \operatorname{ht}(\mathfrak{a}/\mathfrak{p})$$
$$\operatorname{ht}(\mathfrak{a}) = \operatorname{ht}(\mathfrak{a}/\mathfrak{p}) + \operatorname{ht}(\mathfrak{p})$$

*If in addition A is equidimensional (e.g. an integral domain) then it is **biequidimensional** and satisfies for all ideals $\mathfrak{a}$*

$$\dim A = \dim \mathfrak{a} + \operatorname{ht}(\mathfrak{a})$$

In particular for every prime ideal $\mathfrak{p}$ we have

$$\dim A = \dim A_{\mathfrak{p}} + \dim A/\mathfrak{p}$$

and for every maximal ideal

$$\dim A = \dim A_{\mathfrak{m}}$$

Proof. By (3.34.34) A satisfies the criteria in (3.21.10).e) and so is quasi-biequidimensional. The codimension formulas follow from (3.21.12). ■

We may also consider the following result

Proposition 3.34.36 (Generalized Hauptidealsatz for Affine Algebras)
Let A be an affine algebra and $\mathfrak{p}$ a prime ideal. Then

a) If $\operatorname{ht}(\mathfrak{p}) = n$ then it is minimal over some ideal $\mathfrak{a} := (x_1, ..., x_n)$. Furthermore $\mathfrak{a}$ may be chosen such that $\operatorname{ht}(\mathfrak{a}) = n$ and every minimal prime is of height n
b) We have the following characterization of height of a prime ideal

$$\operatorname{ht}(\mathfrak{p}) = \min\{n \mid \mathfrak{p} \text{ minimal over } (x_1, ..., x_n)\}$$

In particular if $\mathfrak{p}$ is minimal over $(x_1, ..., x_n)$ then $\operatorname{ht}(\mathfrak{p}) \leq n$.

Furthermore the same result holds for localization of A at any prime ideal.

Proof. This is largely a restatement of results in the Hauptidealsatz section. By (3.34.32) we have $A/\mathfrak{p}$ is hauptidealsatz for every prime ideal $\mathfrak{p}$. Furthermore by (3.34.35) A is quasi-biequidimensional and so catenary (3.21.10). Then by (3.26.4) this means A is a generalized hauptidealsatz ring (and so is every localization $A_{\mathfrak{p}}$).

Then a) and b) follows from (3.26.12). ■

Corollary 3.34.37
Let A be an affine algebra and $\mathfrak{p}$ a prime ideal. Denote the unique maximal ideal of $A_{\mathfrak{p}}$ by $\mathfrak{m} := \mathfrak{p}A_{\mathfrak{p}}$. Then

$$\dim A_{\mathfrak{p}} = \min\left\{n \;\middle|\; \sqrt{(x_1, ..., x_n)} = \mathfrak{m}\right\} \leq \dim_{k(\mathfrak{m})} \mathfrak{m}/\mathfrak{m}^2$$

with equality iff $A_{\mathfrak{p}}$ is a regular local ring.

Proof. By (3.34.36) $A_{\mathfrak{p}}$ is a Noetherian local ring satisfying the generalized hauptidealsatz. Therefore the result follows by (3.27.2). ■

3.34.5.1 * Biequidimensionality by Strong Normalisation *

We may prove more directly the biequidimensionality property by using the strong form of the Normalisation Lemma (3.34.12) and Going Down (3.23.19). First we prove a technical result

Lemma 3.34.38 (Saturated pairs)
Let $\phi : B \to A$ be an integral map and $\mathfrak{p}_0 \subsetneq \mathfrak{p}_1$ prime ideals lying over $\mathfrak{q}_0 \subsetneq \mathfrak{q}_1$. Then

a) $\mathfrak{q}_0 \subsetneq \mathfrak{q}_1$ saturated $\implies \mathfrak{p}_0 \subsetneq \mathfrak{p}_1$ saturated.
b) $B/\mathfrak{q}_0$ integrally closed and $\mathfrak{p}_0 \subsetneq \mathfrak{p}_1$ saturated $\implies \mathfrak{q}_0 \subsetneq \mathfrak{q}_1$ saturated

We may relax the condition in b) to the existence of another integral map $\psi : C \to B$ such that $C/\psi^{-1}(\mathfrak{q}_0)$ is integrally closed.

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Proof. The first follows by incomparability (3.23.17). The second follows by applying Going Down (3.23.19) to the integral map $B/\mathfrak{q}_0 \hookrightarrow A/\mathfrak{p}_0$. More precisely if $\mathfrak{q}_0 \subsetneq \mathfrak{q} \subsetneq \mathfrak{q}_1$ then $(0) \subsetneq \mathfrak{q}/\mathfrak{q}_0 \subsetneq \mathfrak{q}_1/\mathfrak{q}_0$ whence there exists $\mathfrak{p}$ such that $(0) \subsetneq \mathfrak{p}/\mathfrak{p}_0 \subsetneq \mathfrak{p}_1/\mathfrak{p}_0$. This means $\mathfrak{p}_0 \subsetneq \mathfrak{p} \subsetneq \mathfrak{p}_1$, a contradiction.

The final statement can be demonstrated by applying b) to $C \to A$ and then a) to $B \to A$. ∎

Proposition 3.34.39
*Let A be an affine domain, then every maximal chain has order $n = \dim A$, i.e. A is **irreducible** and **biequidimensional**.*

Proof of (3.34.39). Consider a maximal chain $\mathfrak{p}_0 \subsetneq \mathfrak{p}_1 \subsetneq ... \subsetneq \mathfrak{p}_m$. Clearly $\mathfrak{p}_m$ is maximal by (3.3.37), and as A is integral $\mathfrak{p}_0 = 0$. Apply (3.34.12) to find an integral, injective map

$$\phi : k[X_1, ..., X_n] \hookrightarrow A$$

such that

$$\mathfrak{q}_i := \phi^{-1}(\mathfrak{p}_i) = \left(X_1, ..., X_{h(i)}\right) \quad 0 \leq i \leq m$$

Note $n = \dim A$ by (3.34.28). Clearly $h(0) = 0$ and by (3.23.16) $\mathfrak{q}_m$ is maximal so $h(m) = n$. Observe that

$$k[X_1, ..., X_n]/\mathfrak{q}_i \xrightarrow{\sim} k\left[X_{h(i)+1}, ..., X_n\right]$$

is integrally closed for all i (3.23.12). Therefore we may apply (3.34.38) to ϕ and each pair $\mathfrak{p}_i \subsetneq \mathfrak{p}_{i+1}$, to see that each chain $\mathfrak{q}_i \subsetneq \mathfrak{q}_{i+1}$ is saturated. This can only happen if $h(j) = j$ and therefore $m = n$. ∎

3.34.6 Derivations of Affine Algebras

The notion of derivations is a useful as a coordinate-free construction of the "tangent space" for both differentiable manifolds and algebraic varieties. We review the theory of derivations here primarily focusing on fields and f.g. k-algebras.

Proposition 3.34.40 (Derivations of Polynomial Algebra)
Let $A = k[X_1, ..., X_n]$ be a polynomial algebra and M an (A, B)-bimodule. Then we have an (A, B)-bimodule isomorphism

$$\begin{aligned} \mathrm{Der}_k(A, M) &\cong M^n \\ D &\to (D(X_1), ..., D(X_n)) \\ \sum_{i=1}^n \frac{\partial}{\partial X_i} \cdot v_i &\leftarrow v \end{aligned}$$

When M has a B-basis $\{m_1, ..., m_r\}$ then $\mathrm{Der}_k(A, M)$ also has a B-basis

$$\left\{ \frac{\partial}{\partial X_i} \cdot m_j \right\}_{i=1...n, j=1...r}$$

In particular $\dim_K \mathrm{Der}_k(A, K) = n$.

Proof. The first map is clearly well-defined and by (...) injective. The inverse map is well-defined because we have shown $\frac{\partial}{\partial X_i} \in \mathrm{Der}_k(A, A)$. The chain rule (3.25.8) shows that the maps are mutually inverse (the other direction following simply from orthogonality of the partial derivatives).

The final statement is straightforward. ∎

We may generalise this as follows, so that we may interpret derivations as tangent vectors

Proposition 3.34.41 (Derivations = Tangent Vectors)
Let $A = k[X_1, ..., X_n]/\mathfrak{a}$ be a f.g. k-algebra and M an A-module. Then we have an A-module isomorphism

$$\begin{aligned} \mathrm{Der}_k(A, M) \cong & \left\{ v \in M^n \;\middle|\; \sum_{i=1}^n \overline{\frac{\partial F}{\partial X_i}} v_i = 0 \quad \forall F \in \mathfrak{a} \right\} \subset M^n \\ & D \longrightarrow \left(D\left(\overline{X}_1\right), ..., D\left(\overline{X}_n\right)\right) \\ \sum_{i=1}^n & \overline{\frac{\partial}{\partial X_i}} \cdot v_i \longleftarrow v \end{aligned}$$

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If $\mathfrak{a} = \langle F_1, ..., F_m \rangle$ then this is equal to the kernel of the A-module homomorphism

$$\begin{pmatrix} \overline{\frac{\partial F_1}{\partial X_1}} & \cdots & \overline{\frac{\partial F_1}{\partial X_n}} \\ \vdots & \ddots & \vdots \\ \overline{\frac{\partial F_m}{\partial X_1}} & \cdots & \overline{\frac{\partial F_m}{\partial X_n}} \end{pmatrix} : M^n \to M^m$$

When M is an (A, B)-bimodule then this is also an (A, B)-bimodule isomorphism.

Proof. According to (3.25.8) we have the following relationship for all $F \in k[X_1, ..., X_n]$

$$D(\overline{F}) = D\big(F(\overline{X}_1, ..., \overline{X}_n)\big) = \sum_{i=1}^{n} \frac{\partial F}{\partial X_i}(\overline{X}_i) D(\overline{X}_i) = \sum_{i=1}^{n} \overline{\frac{\partial F}{\partial X_i}} D(\overline{X}_i)$$

When $F \in \mathfrak{a}$ we have $\overline{F} = 0 \implies D(\overline{F}) = 0$, whence the first map is well-defined. Similarly for $v \in \text{RHS}$ we see by definition that

$$\sum_{i=1}^{n} \frac{\partial}{\partial \overline{X_i}} \cdot v_i$$

is zero on $\mathfrak{a}$, so determines a well-defined derivation on A. In the same way we see that the maps are mutually inverse.

Suppose v is in the kernel of the given matrix. For any $G \in \mathfrak{a}$ we have by hypothesis $G = \sum_{j=1}^{m} F_j H_j$ for some $H_j \in k[X_1, ..., X_n]$. Then

$$\begin{aligned} \sum_{i=1}^{n} \overline{\frac{\partial G}{\partial X_i}} v_i &= \sum_{i=1}^{n} \sum_{j=1}^{m} \left(\overline{F_j} \overline{\frac{\partial H_j}{\partial X_i}} + \overline{H_j} \overline{\frac{\partial F_j}{\partial X_i}} \right) v_i \\ &= \sum_{j=1}^{m} \sum_{i=1}^{n} \overline{\frac{\partial F_j}{\partial X_i}} v_i \\ &= 0 \end{aligned}$$

where we have used the fact that $\overline{F_j} = 0$. As G was arbitrary this shows v is in the right-hand side of the isomorphism, and the reverse inclusion is immediate as $F_j \in \mathfrak{a}$. ∎

Proposition 3.34.42 (Lifting for separable algebraic extensions)
Let L/K be a separable algebraic (resp. separably generated) extension (over k) and V an L-module. Then there is an isomorphism (resp. epimorphism) of L-modules

$$\begin{aligned} \text{Der}_k(L, V) &\xrightarrow{\sim} \text{Der}_k(K, V) \\ D &\to D|_K \end{aligned}$$

Proof. Consider $L[V]$ the k-algebra with ideal $N := 0 \times V$ such that $N^2 = 0$ and $L[V]/N \cong L$. Furthermore a derivation $D : K \to V$ corresponds to a unique k-algebra homomorphism $\phi_D : K \to K[V] \hookrightarrow L[V]$ by (3.25.5) so we have the following commutative diagram

$$\begin{array}{ccc} K & \xrightarrow{\phi_D} & L[V] \\ \downarrow & \nearrow & \downarrow \pi_1 \\ L & = & L \end{array}$$

In the algebraic case by (3.25.16) there is a unique homomorphism $\phi_{\tilde{D}} : L \to L[V]$ completing the diagram, and in particular by (3.25.5) there is a derivation $\tilde{D} : L \to V$ extending D. Uniqueness follows from that of $\phi_{\tilde{D}}$.

The separably generated case is similar, except without uniqueness. ∎

Corollary 3.34.43
Let K/k be a separably generated extension, A a K-algebra and M an A-module then

$$\text{Der}_k(A, M) = \text{Der}_K(A, M)$$

Proof. Recall (3.25.1) that a derivation D is K-linear iff it vanishes on K. So we may reduce to the case $A = K$ by considering the restriction $D|_K$.

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Therefore it's sufficient to show that $\mathrm{Der}_k(K,M) = \mathrm{Der}_K(K,M) = 0 = \mathrm{Der}_k(k,M)$, but this follows immediately from (3.34.42). ∎

Proposition 3.34.44 (Transcendence Degree Controls Derivations into Any Module)
*Suppose K/k is **separably generated** (e.g. if k is perfect and K finitely generated), with transcendence basis $\eta_1,...,\eta_n$, $n = \mathrm{trdeg}_k(K)$, and let M be a K-module. Then*

$$\mathrm{Der}_k(K,M) \cong M^n, \quad D \mapsto (D(\eta_1),...,D(\eta_n))$$

as K-modules. In particular, taking $M := K$, $\mathrm{trdeg}_k(K) = \dim_K \mathrm{Der}_k(K)$.

Proof. By hypothesis $K/k(\eta_1,...,\eta_n)$ is algebraic and separable. Therefore we have

$$\mathrm{Der}_k(K,M) \overset{(3.34.42)}{\cong} \mathrm{Der}_k(k(\eta_1,...,\eta_n),M) \overset{(3.25.10)}{\cong} \mathrm{Der}_k(k[\eta_1,...,\eta_n],M) \overset{(3.34.40)}{\cong} M^n$$

the last isomorphism sending $D \mapsto (D(\eta_1),...,D(\eta_n))$, matching the same map once restricted back along the two previous isomorphisms (both restriction maps, $D \mapsto D|_{k(\eta_1,...,\eta_n)}$ and $D \mapsto D|_{k[\eta_1,...,\eta_n]}$, fix the value at each η_i). Taking $M := K$ gives $\dim_K \mathrm{Der}_k(K,K) = n = \mathrm{trdeg}_k(K)$. ∎

Proposition 3.34.45 (Derivations Extend Uniquely Along a Field Extension)
Let K/k be a field extension, k'/k any field extension, and $D \in \mathrm{Der}_k(K,K)$. There is a unique $\tilde{D} \in \mathrm{Der}_{k'}(K \otimes_k k', K \otimes_k k')$ with $\tilde{D}(x \otimes 1) = D(x) \otimes 1$ for every $x \in K$; explicitly, $\tilde{D}(x \otimes c) = D(x) \otimes c$.

Proof. Drafted by Claude Sonnet, reviewed by David Rufino.

Existence. Define $\tilde{D}(x \otimes c) := D(x) \otimes c$, extended k'-linearly to $K \otimes_k k'$ (well-defined as D is k-linear). It satisfies the Leibniz rule on simple tensors, $\tilde{D}((x \otimes c)(y \otimes c')) = D(xy) \otimes cc' = (D(x)y + xD(y)) \otimes cc' = \tilde{D}(x \otimes c) \cdot (y \otimes c') + (x \otimes c) \cdot \tilde{D}(y \otimes c')$, hence on all of $K \otimes_k k'$ by bilinearity: $\tilde{D} \in \mathrm{Der}_{k'}(K \otimes_k k', K \otimes_k k')$.

Uniqueness. $K \otimes_k k'$ is k'-spanned by $K \otimes 1$ (extension of scalars along a k-basis of K), and any extension of D is pinned down there ($x \otimes 1 \mapsto D(x) \otimes 1$ by definition of "extends"), so agrees with $\tilde{D}$ on a k'-spanning set, hence everywhere by k'-linearity. ∎

Lemma 3.34.46
Let A be a k-algebra and $\mathfrak{m}$ an ideal. Then $\mathfrak{m}/\mathfrak{m}^2$ is a $k(\mathfrak{m})$-module with action given by

$$(a + \mathfrak{m}) \cdot (b + \mathfrak{m}^2) = (ab + \mathfrak{m}^2)$$

for all $a \in A$ and $b \in \mathfrak{m}$.

Proof. Suppose $a - a' \in \mathfrak{m}$ and $b - b' \in \mathfrak{m}^2$ then

$$ab - a'b' = (a - a')(b - b') + a'(b - b') + b'(a - a') \in \mathfrak{m}^2$$

this shows that the action is well-defined. The properties of a module are straightforward. ∎

Lemma 3.34.47
Let A be a k-algebra and $\mathfrak{m}$ a maximal ideal. Then there is an isomorphism of A-algebras

$$\begin{array}{ccc} A & \longrightarrow & A/\mathfrak{m}^2 \\ \downarrow{\scriptstyle \pi_1} & \overset{\pi_2}{\searrow} & \downarrow \\ k(\mathfrak{m}) & \overset{\psi}{\underset{\sim}{\dashrightarrow}} & k(\mathfrak{m}/\mathfrak{m}^2) \end{array}$$

Further there is an isomorphism of A-modules

$$\begin{aligned} \mathrm{Der}_k(A/\mathfrak{m}^2, k(\mathfrak{m}/\mathfrak{m}^2)) &\to \mathrm{Der}_k(A, k(\mathfrak{m})) \\ D &\to x \to \psi^{-1}(D(\overline{x})) \end{aligned}$$

Proof. The first statement follows from (3.3.58) with $\mathfrak{a} = \mathfrak{m}^2$ and $\mathfrak{b} = \mathfrak{m}$.

Define $\hat{D}(x) := \psi^{-1}(D(\overline{x}))$. By hypothesis $D(\overline{xy}) = \pi_2(x)D(\overline{y}) + \pi_2(y)D(\overline{x})$. Therefore

$$\hat{D}(xy) = \pi_1(x)\psi^{-1}(D(\overline{y})) + \pi_1(y)\psi^{-1}(D(\overline{x})) = \pi_1(x)\hat{D}(\overline{y}) + \pi_1(y)\hat{D}(\overline{x})$$

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and so $\hat{D}$ is a well-defined derivation. Suppose $\hat{D} \in \mathrm{Der}_k(A, k(\mathfrak{m}))$ then by the product rule $\mathfrak{m}^2 \subset \ker\left(\hat{D}\right)$ so there is a derivation $D' : A/\mathfrak{m}^2 \to k(\mathfrak{m})$ such that $D'(\overline{x}) = \hat{D}(x)$. Define $D \coloneqq \psi \circ D'$, then evidentally the image of this derivation is $\hat{D}$ and so the map is surjective.

Suppose $\hat{D} = 0$ then as ψ is an isomorphism $D \circ \bar{\cdot} = 0$. As the reduction map is surjective we deduce that $D = 0$. Therefore the map is injective, bijective and an isomorphism. ∎

Lemma 3.34.48
Let A be a k-algebra and $\mathfrak{m}$ an ideal. Then there is an isomorphism of A-modules

$$\begin{aligned}\mathrm{Hom}_{k(\mathfrak{m})}(\mathfrak{m}/\mathfrak{m}^2, k(\mathfrak{m})) &\to \mathrm{Hom}_{k(\mathfrak{m}/\mathfrak{m}^2)}((\mathfrak{m}/\mathfrak{m}^2)/(\mathfrak{m}^2/\mathfrak{m}^2), k(\mathfrak{m}/\mathfrak{m}^2)) \\ \theta &\to \psi \circ \theta \circ i^{-1}\end{aligned}$$

where $i : \mathfrak{m}/\mathfrak{m}^2 \to (\mathfrak{m}/\mathfrak{m}^2)/(\mathfrak{m}^2/\mathfrak{m}^2)$ is a ring isomorphism such that

$$i(\pi_1(a) \cdot x) = \pi_2(a) \cdot i(x)$$

Proof. Then for $x \in (\mathfrak{m}/\mathfrak{m}^2)/(\mathfrak{m}^2/\mathfrak{m}^2)$, $a \in A$ and $\theta \in \mathrm{Hom}_{k(\mathfrak{m})}(\mathfrak{m}/\mathfrak{m}^2, k(\mathfrak{m}))$

$$\begin{aligned}(\psi \circ \theta \circ i^{-1})(\pi_2(a) \cdot x) &= (\psi \circ \theta)(\pi_1(a) \cdot i^{-1}(x)) \\ &= \psi(\pi_1(a)\theta(i^{-1}(x))) \\ &= \pi_2(a)(\psi \circ \theta \circ i^{-1})(x)\end{aligned}$$

so that the map is well-defined. Similarly if $\theta' \in \mathrm{Hom}_{k(\mathfrak{m}/\mathfrak{m}^2)}((\mathfrak{m}/\mathfrak{m}^2)/(\mathfrak{m}^2/\mathfrak{m}^2), k(\mathfrak{m}/\mathfrak{m}^2))$

$$\begin{aligned}(\psi^{-1} \circ \theta' \circ i)(\pi_1(a) \cdot x) &= (\psi^{-1} \circ \theta')(\pi_2(a) \cdot i(x)) \\ &= \psi^{-1}(\pi_2(a)\theta'(i(x))) \\ &= \pi_1(a)(\psi^{-1} \circ \theta' \circ i)(x)\end{aligned}$$

so the inverse map is well-defined. ∎

Proposition 3.34.49 (Coefficient Field mod $\mathfrak{m}^2$)
Let A be a k-algebra and $\mathfrak{m} \triangleleft A$ a maximal ideal with $\mathfrak{m}^2 = 0$ and residue field $k(\mathfrak{m}) \coloneqq A/\mathfrak{m}$. If $k(\mathfrak{m})/k$ is separably generated (e.g. if k is perfect and $k(\mathfrak{m})/k$ is finitely generated), then the quotient map $\pi : A \to k(\mathfrak{m})$ admits a k-algebra section $j : k(\mathfrak{m}) \to A$, that is there is a subfield $k \subseteq \hat{k} \subseteq A$ such that $\pi|_{\hat{k}} : \hat{k} \to k(\mathfrak{m})$ is an isomorphism. In particular $A = \hat{k} \oplus \mathfrak{m}$ as k-vector spaces.

Proof. Drafted by Claude Sonnet, reviewed by David Rufino.

As $\mathfrak{m}^2 = 0$, $\mathfrak{m}$ is the only prime of A, so A is local, and $A/\mathfrak{m}^2 = A$; so (3.25.18) a) (with $n = 1$) gives a k-algebra homomorphism $j \coloneqq s_1 : k(\mathfrak{m}) \to A$ with $\pi \circ j = \mathrm{id}_{k(\mathfrak{m})}$.

Set $\hat{k} \coloneqq j(k(\mathfrak{m})) \subseteq A$. As j is a ring homomorphism from a field, it is injective (3.3.61), so $j : k(\mathfrak{m}) \to \hat{k}$ is an isomorphism, and $\pi \circ j = \mathrm{id}$ shows $\pi|_{\hat{k}}$ is its inverse. Finally j is a k-linear section of the k-linear surjection $\pi : A \to k(\mathfrak{m})$ with kernel $\mathfrak{m}$, so $A = \mathrm{Im}(j) \oplus \ker(\pi) = \hat{k} \oplus \mathfrak{m}$ as k-vector spaces. ∎

Proposition 3.34.50 (Tangent space is dual to Cotangent space)
Let A be a k-algebra and $\mathfrak{m} \triangleleft A$ a maximal ideal with residue field $k(\mathfrak{m}) \coloneqq A/\mathfrak{m}$. There is a homomorphism of $k(\mathfrak{m})$-modules

$$\begin{aligned}\mathrm{Der}_k(A, k(\mathfrak{m})) &\longrightarrow \mathrm{Hom}_{k(\mathfrak{m})}(\mathfrak{m}/\mathfrak{m}^2, k(\mathfrak{m})) \\ D &\longrightarrow \theta_D : \overline{x} \to D(x)\end{aligned}$$

*When $k(\mathfrak{m})/k$ is **separably generated** then this is an isomorphism. In particular this holds when k is perfect and $k(\mathfrak{m})/k$ is finitely generated.*

Proof. The map θ_D is well-defined because D annihilates $\mathfrak{m}^2$ by the product rule. It remains to show the map is bijective when $k(\mathfrak{m})/k$ is separably generated.

We first claim we can reduce to the case $\mathfrak{m}^2 = 0$. For consider $A' \coloneqq A/\mathfrak{m}^2$ and $\mathfrak{m}' = \mathfrak{m}/\mathfrak{m}^2$ and the commutative diagram of A-module homomorphisms

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$$\begin{array}{ccc} \mathrm{Der}_k(A', k(\mathfrak{m}')) & \longrightarrow & \mathrm{Hom}_{k(\mathfrak{m}')}\left(\mathfrak{m}'/(\mathfrak{m}')^2, k(\mathfrak{m}')\right) \\ \downarrow \sim & & \downarrow \sim \\ \mathrm{Der}_k(A, k(\mathfrak{m})) & \longrightarrow & \mathrm{Hom}_{k(\mathfrak{m})}(\mathfrak{m}/\mathfrak{m}^2, k(\mathfrak{m})) \end{array}$$

The horizontal maps have already been defined and the vertical maps are defined in (3.34.47) and (3.34.48). The diagram evidentally commutes, so therefore it is sufficient to show that the top map is an isomorphism.

We revert to the original notation, but with the additional assumption that $\mathfrak{m}^2 = 0$. By (3.34.49) there is a subfield $k \subseteq \hat{k} \subseteq A$ mapping isomorphically to $k(\mathfrak{m})$ under the canonical projection $\pi : A \to A/\mathfrak{m} = k(\mathfrak{m})$, giving a direct sum of k-vector spaces

$$A = \hat{k} \oplus \mathfrak{m}$$

Given $\theta \in \mathrm{Hom}_{k(\mathfrak{m})}(\mathfrak{m}, k(\mathfrak{m}))$ define

$$D_\theta(\lambda + x) := \theta(x) \quad \lambda \in \hat{k}, x \in \mathfrak{m}$$

It satisfies the product rule for

$$\begin{aligned} D_\theta((\lambda + x)(\lambda' + x')) &= \theta(\lambda' x + \lambda x') \\ &= \pi(\lambda')\theta(x) + \pi(\lambda)\theta(x') \\ &= \pi(\lambda' + x')\theta(x) + \pi(\lambda + x)\theta(x') \\ &= (\lambda' + x') \cdot D_\theta(\lambda + x) + (\lambda + x) \cdot D_\theta(\lambda' + x') \end{aligned}$$

Therefore the given map is surjective.

As $k(\mathfrak{m})/k$ is separably generated then by (3.34.43) any derivation D is $\hat{k}$-linear. Therefore

$$D(\lambda + x) = D|_{\mathfrak{m}}(x) \quad \lambda \in \hat{k}, x \in \mathfrak{m}$$

which shows that the given map is injective. ∎

Remark 3.34.51
The proof is substantially simpler when $k(\mathfrak{m}) = k$, for example when k is algebraically closed.

The argument follows the suggestion of [Har13] (Ex 8.1), but using the simpler result [Mat70] (Prop 28.I) to argue more directly.

3.34.7 Linearly Disjoint Algebras

Definition 3.34.52
*Let A/k, B/k be k-algebras and homomorphisms $i : A/k \to \Omega/k$, $j : B/k \to \Omega/k$ (typically being inclusion). Then we say that A and B are **linearly disjoint in Ω** if the canonical map*

$$\begin{aligned} A \otimes_k B &\to \Omega \\ a \otimes b &\to i(a)j(b) \end{aligned}$$

is injective. As the canonical maps $A, B \to A \otimes_k B$ are injective by (3.4.41) this implies i, j must be injective. In the case that Ω is integral (resp. reduced) then A, B are also necessarily integral (resp. reduced).

Recall that every k-module is free (3.3.126), moreover by (3.4.23) a basis of $A \otimes_k B$ may be formed by tensor product of bases for A and B.

Example 3.34.53
Let K/k be a non-trivial field extension then K is not linearly disjoint with itself. For $\{1, x\}$ a linearly independent subset of K which may be extended to a basis. Given the remark on the basis of the tensor product we see $1 \otimes x \neq x \otimes 1$. On the other hand the image of $1 \otimes x - x \otimes 1$ in Ω is clearly 0. The same consideration shows that $A \cap B = k$.

Remark 3.34.54
The notion of linear disjointness in general depends on the embeddings in Ω/k. For example $k(t)$ and $k(s)$ are linearly disjoint in $k(s,t)$, but not when both are identified with $k(x)$.

Proposition 3.34.55
Let $i : A/k \to \Omega/k$ and $j : B/k \to \Omega/k$ be algebra homomorphisms. Then the following are equivalent

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- ***a)*** *A/k and B/k are linearly disjoint in Ω/k*
- ***b)*** *If $S \subset A$ is linearly independent over k then $i(S)$ is linearly independent over B*
- ***b')*** *If $T \subset B$ is linearly independent over k then $j(T)$ is linearly independent over A*
- ***c)*** *There is a k-basis $\{a_\lambda\}_{\lambda\in\Lambda}$ of A for which $\{i(a_\lambda)\}_{\lambda\in\Lambda}$ is linearly independent over B*
- ***c')*** *There is a k-basis $\{b_\lambda\}_{\lambda\in\Lambda}$ of B for which $\{j(b_\lambda)\}_{\lambda\in\Lambda}$ is linearly independent over A*

Proof. $a) \implies b)$ By (3.4.30) and (3.4.31) there is an injective map

$$B^{(S)} \hookrightarrow A \otimes_k B$$
$$(b_s)_{s\in S} \to \sum_{s\in S} s \otimes b_s$$

By hypothesis the composite with $A \otimes_k B \hookrightarrow \Omega$ is injective which is precisely the required conclusion.

$b) \implies c)$ By (3.3.126) there exists a basis which by hypothesis is linearly independent over B

$c) \implies a)$ By (3.4.23) there is an isomorphism

$$B^{(\Lambda)} \cong A \otimes_k B$$

and the canonical map into Ω is identified with $(b_\lambda)_{\lambda\in\Lambda} \to \sum i(a_\lambda)j(b_\lambda)$. The hypothesis means precisely that this map is injective.

We immediately have the symmetric result $a) \iff b') \iff c')$. ∎

In cases where both algebras are fields and one is algebraic then linear disjointness can be characterized by a property of the tensor product. In particular it does not depend on the **ambient** algebra Ω/k.

Lemma 3.34.56
Let A be a finite-dimensional k-algebra with no zero-divisors. Then A is a field.

Proof. By assumption multiplication by a non-zero element x is injective and k-linear. Therefore by (3.3.137) it is surjective and x has an inverse as required. ∎

Proposition 3.34.57
*Let K/k be a field extension and k'/k an **algebraic** field extension. Then the following are equivalent*

a) K and k' are linearly disjoint with respect to every embedding into an algebra Ω/k
b) K and k' are linearly disjoint with respect to some common field extension Ω/k
c) $K \otimes_k k'$ is an integral domain
d) $K \otimes_k k'$ is a field

Proof. $a) \implies b)$ We may take $\Omega := (K \otimes_k k')/\mathfrak{m}$.

$b) \implies c)$ As Ω is assumed to be a field it is certainly integral, and therefore so is $K \otimes_k k'$ being a subring.

$c) \implies d)$ As every element of the tensor product is a finite linear combination of elementary tensors we may reduce to the case k' is finitely generated. Then by (3.18.44) k' is finite over k and therefore by (3.4.23) $K \otimes_k k'$ is finite over K. By (3.34.56) it is a field.

$d) \implies a)$ This is immediate because the kernel is an ideal which must be (0). ∎

3.34.8 Finiteness of Integral Closure

Remark 3.34.58
In what follows we are interested in the integral closure of affine domains in finite extensions of the quotient field. Similar results are obtained for Dedekind Domains (3.31.11) with the restriction that the field extension is separable. Although this holds over number fields where the characteristic is zero, this will not in general be the case for varieties over finite fields, as for example $\mathbb{F}_p(t)$ admits the inseparable extension $\mathbb{F}_p(t^{1/p})$. However the finiteness still holds, because it's easy to see that the integral closure of $\mathbb{F}_p[t]$ is simply $\mathbb{F}_p[t^{1/p}]$. This generalises because every purely inseparable extension is contained in an extension obtained by adjoining p-th roots of unity.

Proposition 3.34.59 (Normalisation over Finite Fields)
Let k be a (possibly) finite field and $A = k[x_1, ..., x_n]$ be a finitely-generated k-algebra. Then there exists a normalising family $z_1, ..., z_d \in A$ such that $k[x_1, ..., x_n]$ is integral over $k[z_1, ..., z_d]$.

Proof. TODO ∎

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Proposition 3.34.60 (Affine Domain are Japanese Rings)
Let A be an affine domain over k with $K = \mathrm{Frac}(A)$ and L/K a finite extension. Then $\overline{A}^L$ is a finite A-module and Noetherian with quotient field L.

Furthermore $\dim \overline{A}^L = \dim A$.

Proof. Suppose $A = k[x_1, ...x_n]$, apply the normalisation lemma (3.34.59) to obtain a lattice of subrings

$$\begin{array}{c} L \\ | \\ K \\ | \\ A \\ | \\ A' := k[z_1, ..., z_d] \end{array}$$

where $z_1, ..., z_d$ are algebraically independent over k. We claim it is enough to show that the $\overline{A'}^L$ is a finite A'-module. For A is finite and integral over A'. Therefore by (3.23.28) $\overline{A}^L = \overline{A'}^L$ is finite over A', and a-fortiori over A.

Therefore without loss of generality we may assume that $x_1, ..., x_n$ are algebraically independent over k and A is integrally closed (3.23.12).

By (3.18.73) we may find a finite extension L'/L such that L'/K is normal. So by (3.23.29) we may assume without loss of generality that L/K is normal.

By (...) we may decompose this into a tower $M/L/K$ such that M/L is Galois and L/K is purely inseparable.

We first show that the integral closure of A in L is a finite A-module. When $L = K$ this is trivial because A is already integrally closed in its quotient field K. Therefore we may assume $L \neq K$ and the characteristic p is non-zero (3.18.103). $L = K(y_1, ..., y_n)$ is finitely generated and by definition $y_i^q \in K$ for some p-th power q. Therefore L is contained in $L' := k\left(x_1^{1/q}, ..., x_n^{1/q}\right)$ which is again a finite purely inseparable extension of K. The elements $x_1^{1/q}, ..., x_n^{1/q}$ are again algebraically independent so that the ring $A' := k\left[x_1^{1/q}, ..., x_n^{1/q}\right]$ is integrally closed (3.23.12) with quotient field L'. Evidently A' is finite and integral over A (3.23.4) therefore (3.23.28) $\overline{A}^{L'} = \overline{A'}^{L'} = A'$ is finite over A. By (3.23.29) then $\overline{A}^L$ is finite over A and $\overline{A}^L$ has quotient field L (3.23.24). As A is Noetherian then so is $\overline{A}^L$.

By transitivity we may then reduce to the case L/K is separable which was proven in (3.23.26). ∎

Corollary 3.34.61
Let A be an affine domain of Krull dimension 1, with quotient field K and L/K a finite extension. Then the integral closure $\overline{A}^L$ is a finite A-module and a Dedekind domain.

Further for every prime ideal $\mathfrak{p}$ of A we have the ramification formula

$$\sum_{\mathfrak{B}|\mathfrak{p}} e_{\mathfrak{B}/\mathfrak{p}} f_{\mathfrak{B}/\mathfrak{p}} = [L : K]$$

Proof. By (3.34.60) $B := \overline{A}^L$ is a finite A-module with quotient field L and Noetherian. By (...) it is integrally closed and by (3.23.22) has Krull Dimension 1. Therefore by definition B is also a Dedekind domain.

The ramification formula follows from (3.31.20). ∎

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